

ECE264D:

CMOS Analog Integrated Circuits and Systems IV

Lecture 11:

All Digital Phase Locked Loops (ADPLL)

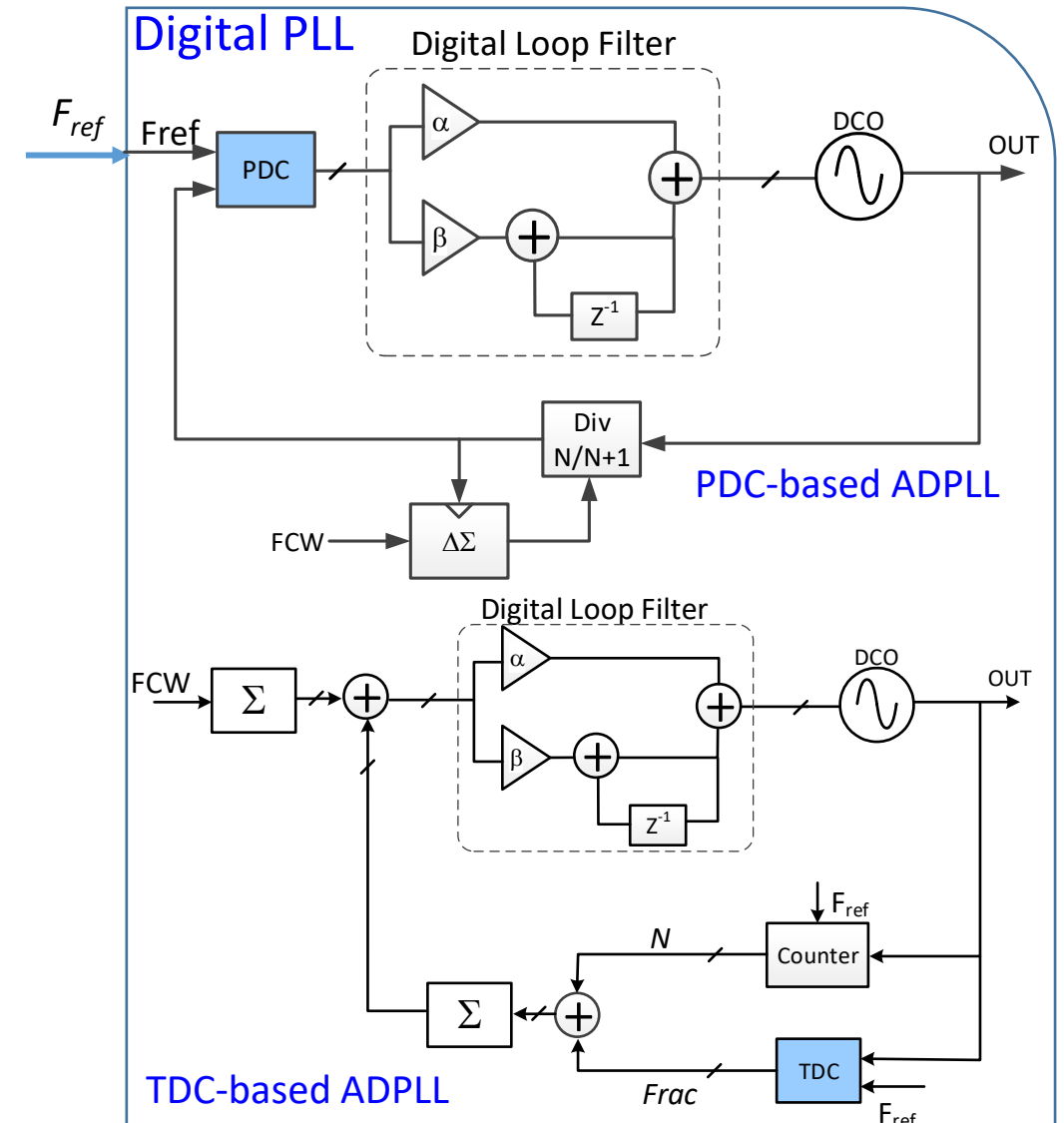
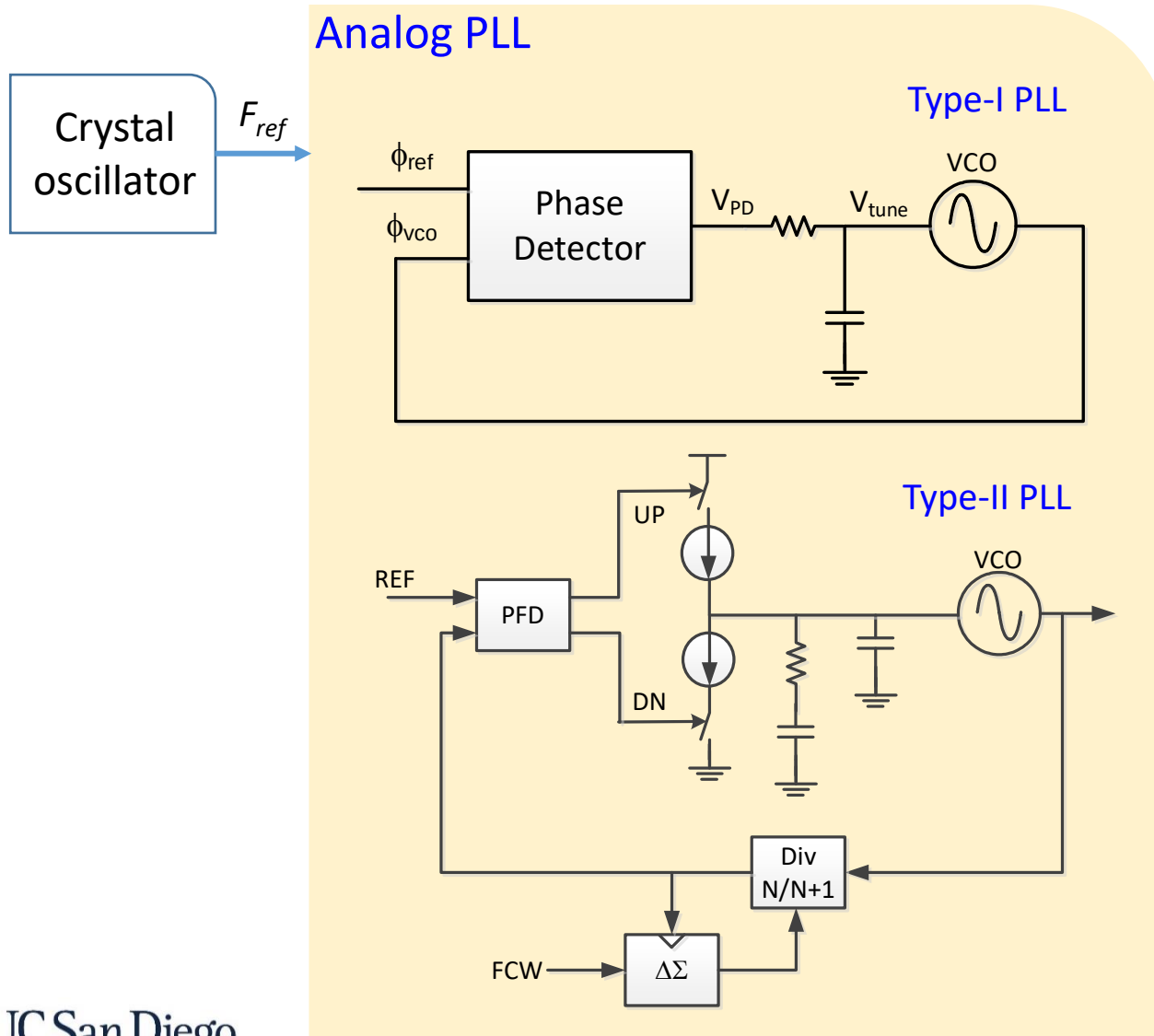
TDC-Based and PDC-Based ADPLL

Time to Digital Conversion (TDC)

Yiwu Tang



Frequency Synthesizer Overview

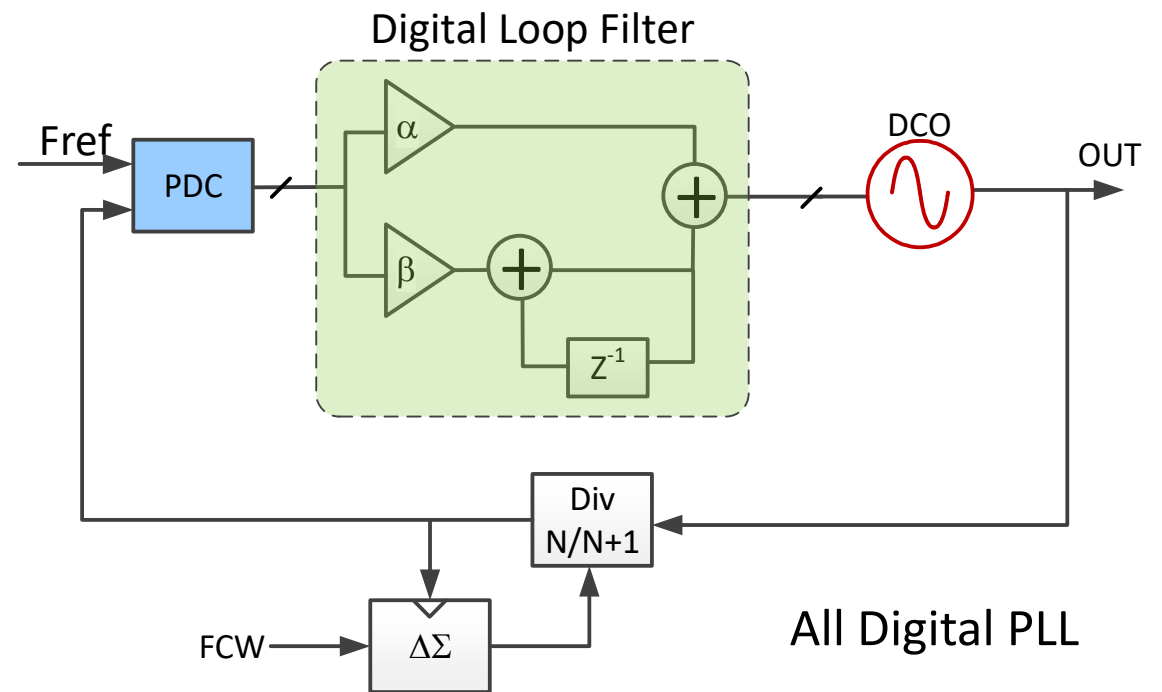
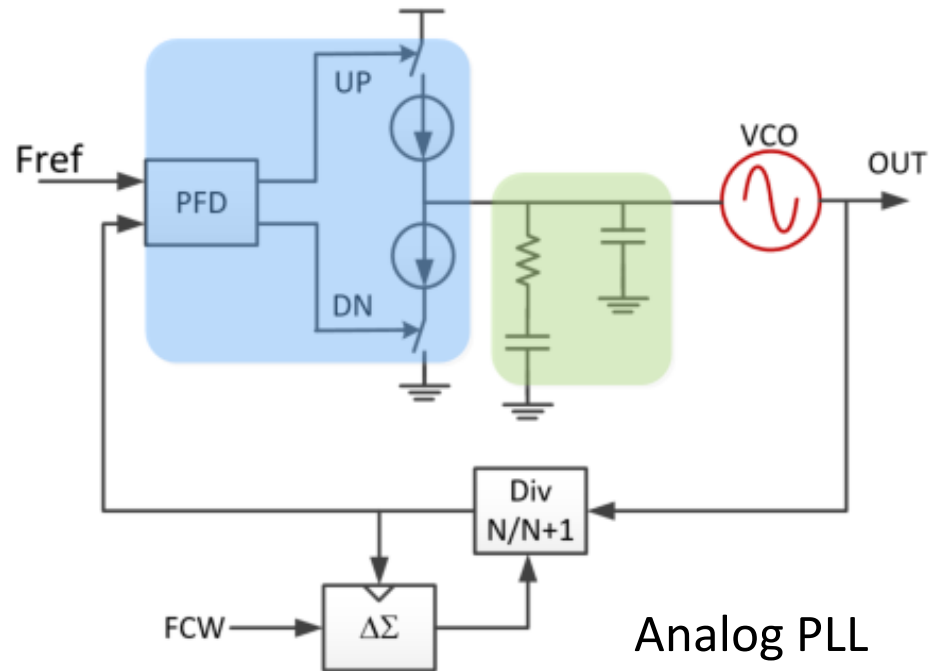


Outline

- All digital phase locked loop (ADPLL) architecture
 - Comparison with Analog PLL
- Two types of ADPLL
 - TDC-ADPLL
 - PDC-ADPLL
- Time to digital converters (TDC)
 - TDC topologies
 - TDC performance specifications
 - TDC quantization noise



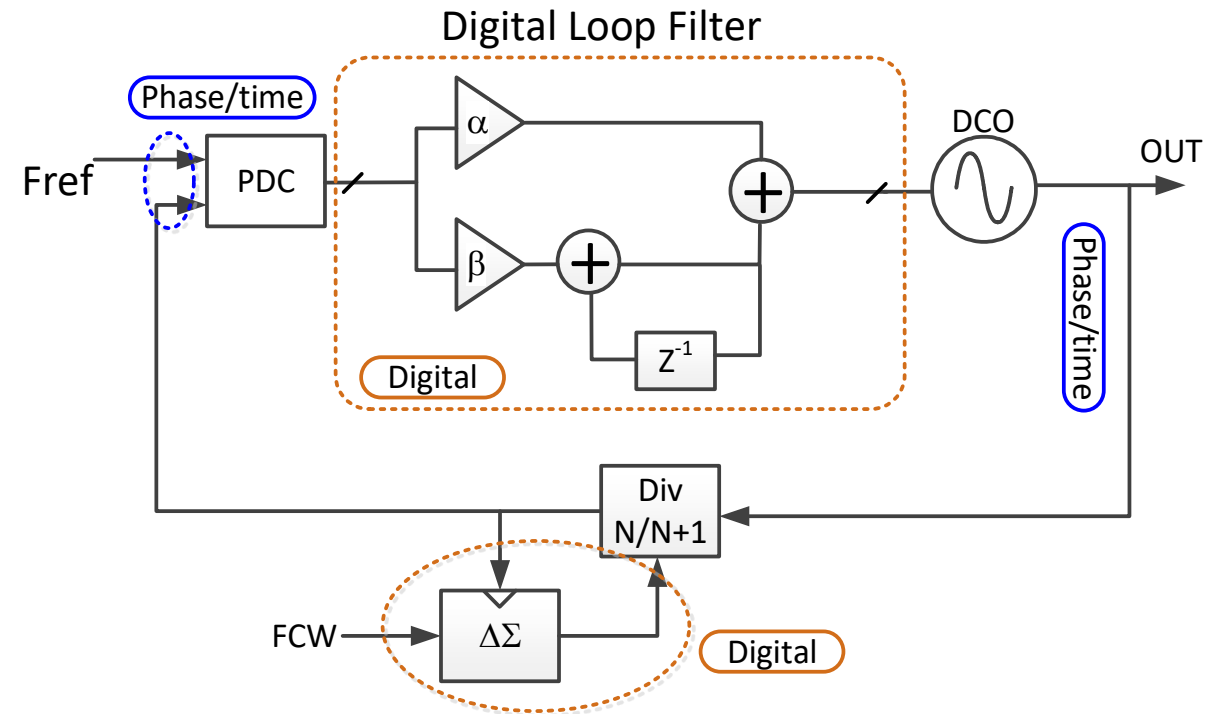
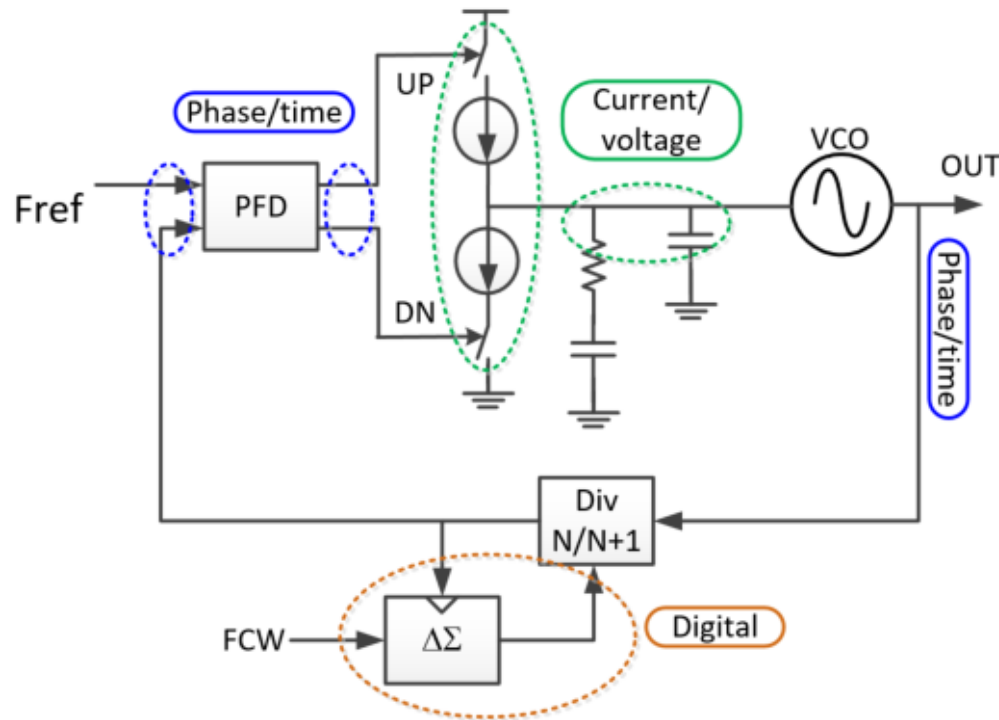
Analog PLL $\Rightarrow$ Digital PLL



- PFD and charge pump $\Rightarrow$ Phase to digital converter (PDC)
- RC loop filter $\Rightarrow$ Digital IIR filter
- Voltage controlled oscillator (VCO) $\Rightarrow$ digitally controlled oscillator (DCO)



What is ADPLL (All Digital PLL)?



- Building blocks
 - Digital blocks (such as digital loop filter and DSM)
 - Digital-analog/RF mixed signal blocks (such as PDC and DCO)
- Signal processing around the loop
 - Mainly in digital domain instead of voltage/current/charge domain
 - Phase/timing signals are still the primary signal form



Why ADPLL

- Disadvantages of analog PLL
 - Analog loop filter: large area or off-chip capacitance if loop bandwidth is narrow; inflexible in loop bandwidth programmability or adaptation (gear-shifting); loop filter leakage
 - Charge pump headroom reduces as process node evolves to nanometer with lower and lower VDD voltage
 - On the other hand, VCO requires large voltage range for varactor tuning
 - Less convenient to perform digital phase modulation (e.g. GSM/EDGE phase modulation)
 - Less convenient to perform digital signal processing, such as DSM quantization noise cancellation.
- There are many advantages in analog PLL: robust, superior in-band phase noise, low spur, low power – still the choice of many high performance frequency synthesizers.



Why ADPLL

- What does ADPLL offer
 - Digital loop filter: small area, no off-chip capacitance; flexible in loop bandwidth programmability or adaptation (gear-shifting), RTL design and digital synthesis, easy to port from one process node to another.
 - DCO control by digital bits from digital loop filter without voltage range constraints and more compatible with low VDD supply
 - Convenient to perform digital phase modulation (e.g. GSM/EDGE phase modulation)
 - Convenient to perform digital signal processing, such as DSM quantization noise cancellation.
- ADPLL is not exactly “all” digital. DCO is RF design with digital control, PDC/TDC is analog intensive mixed signal design. They are not easy to port from one process node to another and not digital process friendly.

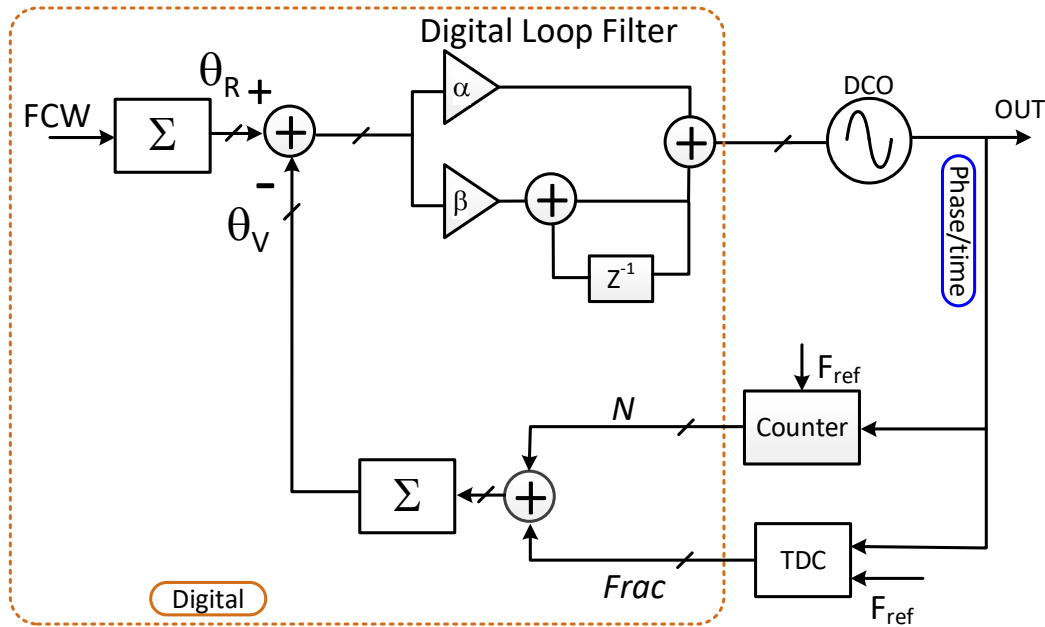


Challenges of ADPLL

- Quantization noise
 - PLL processes phase/voltage signal in analog domain. ADPLL partially processes the signal in the digital domain. Each time analog to digital (TDC) or digital to analog (DCO) occurs, there is quantization noise introduced that contributes to phase noise.
 - With process scaling, gate delay is reducing, so is the TDC quantization noise. For example, TDC resolution limited to one gate delay goes from 10~20ps in 65nm to 5~10ps in 28nm and continues improving.
- Digital spurs and noise
 - DCO is digitally controlled with many toggling digital lines routed to the oscillator core. The oscillator is prone to digital spurs and noise.

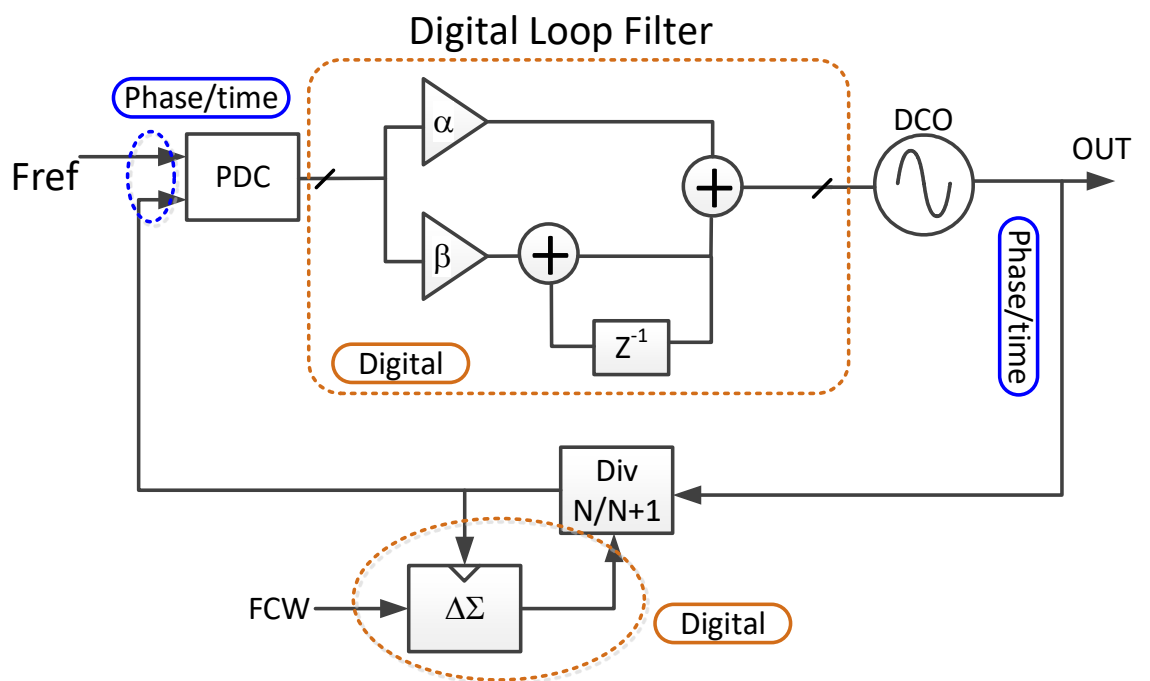


Two Types of ADPLL



TDC-ADPLL

- Direct phase domain operation, processing absolute phase of DCO
- TDC in feedback path measuring absolute time, need gain calibration, why?
- No DSM needed $\Rightarrow$ No DSM Qn, allows wide LBW
- Counter and TDC operate at DCO frequency – high power consumption



PDC-ADPLL

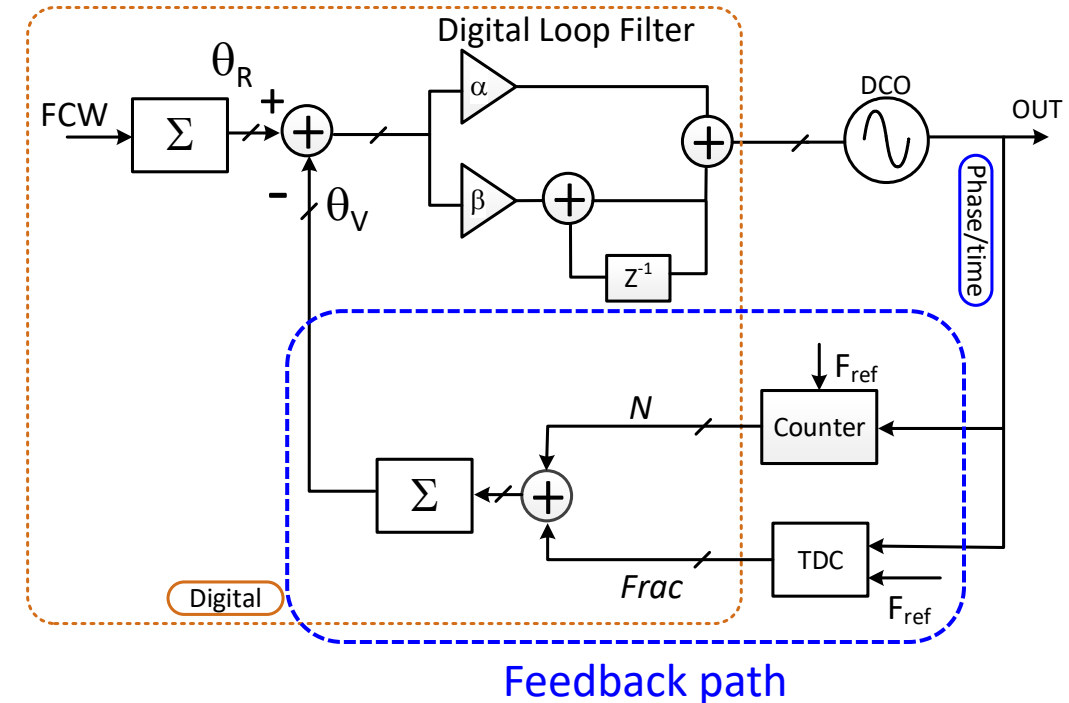
- PDC in feedforward path measures relative phase and replaces PFD in analog PLL
- Similar to APLL that uses N-Div. Quantize F_{ref} and F_v phase difference and process digitally instead of using PFD/CP/LF in charge/voltage domain.



TDC-ADPLL

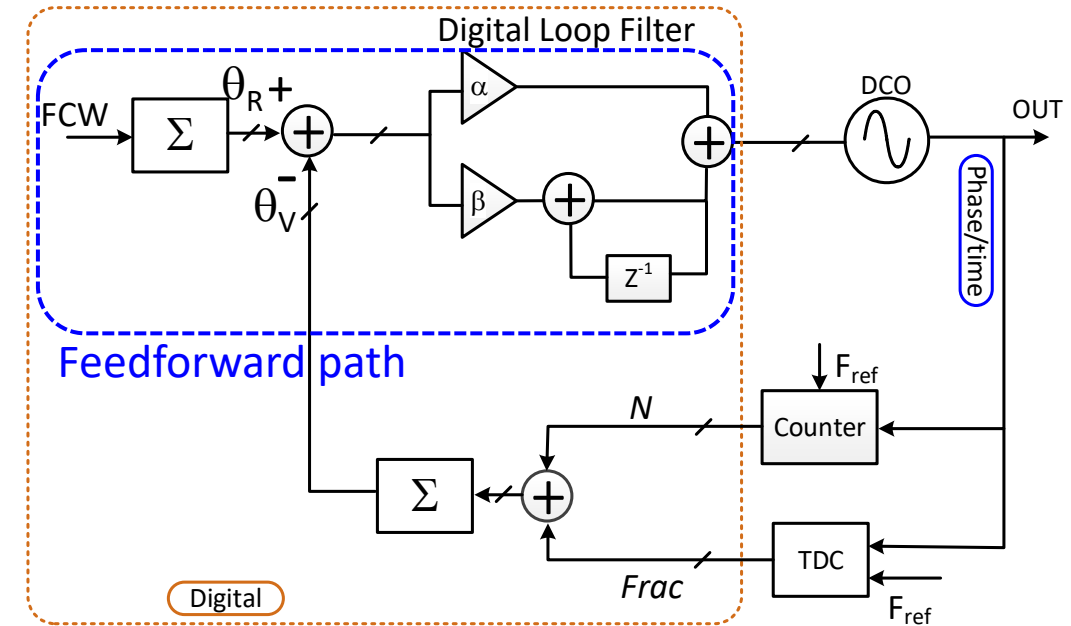
- Feedback path

- Counter counts the number of DCO periods in one reference clock period $\Rightarrow N$
- TDC quantizes the remaining fractional DCO period $T_{\text{ref}} - NT_{\text{DCO}} \Rightarrow \text{Frac}$
- In the steady state, $N + \text{Frac} = N \cdot \alpha$, where $N \cdot \alpha$ is represented by FCW
- TDC acts as a ruler with LSB time as the ruler unit. For $N + \text{Frac}$ to be meaningful in calculation, TDC LSB unit time and DCO period need to be a known ratio. TDC is often implemented as a delay chain and the delay of each cell is the LSB unit time. It is highly PVT dependent and requires calibration using DCO period. It measures the absolute time $\Rightarrow$ **TDC**
- Accumulation of $N + \text{Frac}$, $\sum_1^k (N + \text{Frac})$, represents DCO phase θ_v with DCO period as unit



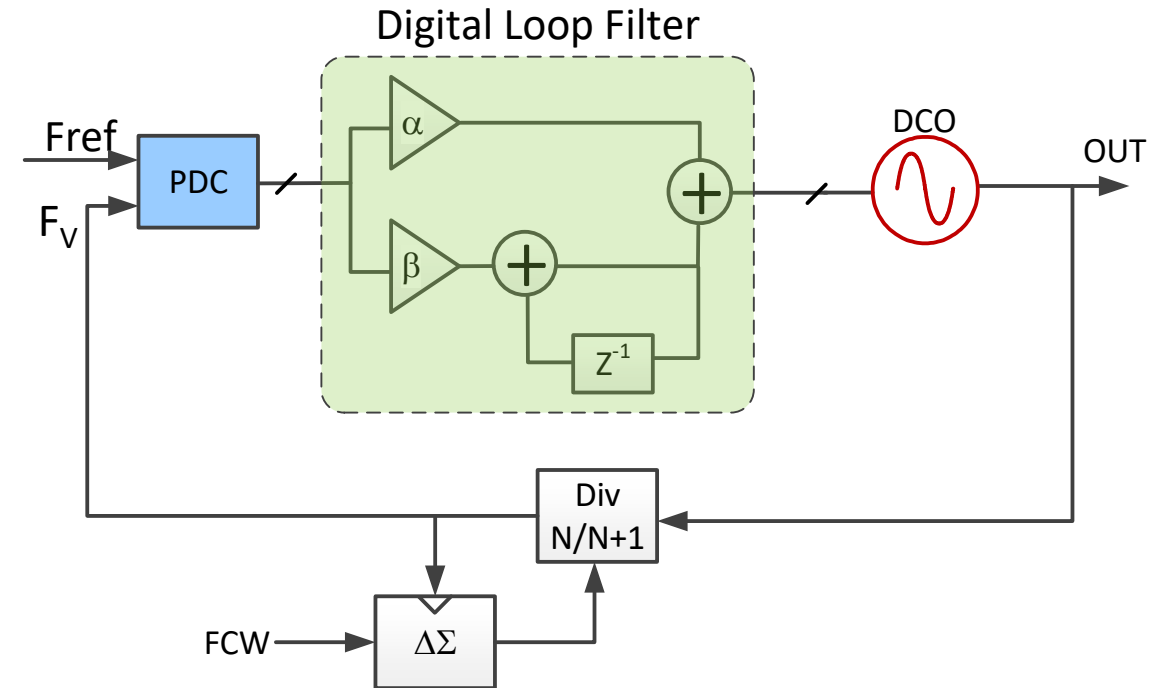
TDC-ADPLL

- FCW: frequency command word
 - Digital representation of divide ratio $N \cdot \alpha$
 - Need enough bit-width for phase/frequency accuracy
- Feedforward path
 - During each reference period, reference phase θ_R increases by FCW (ignoring 2π factor for simplification) $\theta_R = \sum_{i=1}^k FCW$
 - The difference of reference phase and DCO phase, $\theta_R - \theta_V$, is filtered by digital loop filter.
 - The filter output controls DCO to adjust frequency.



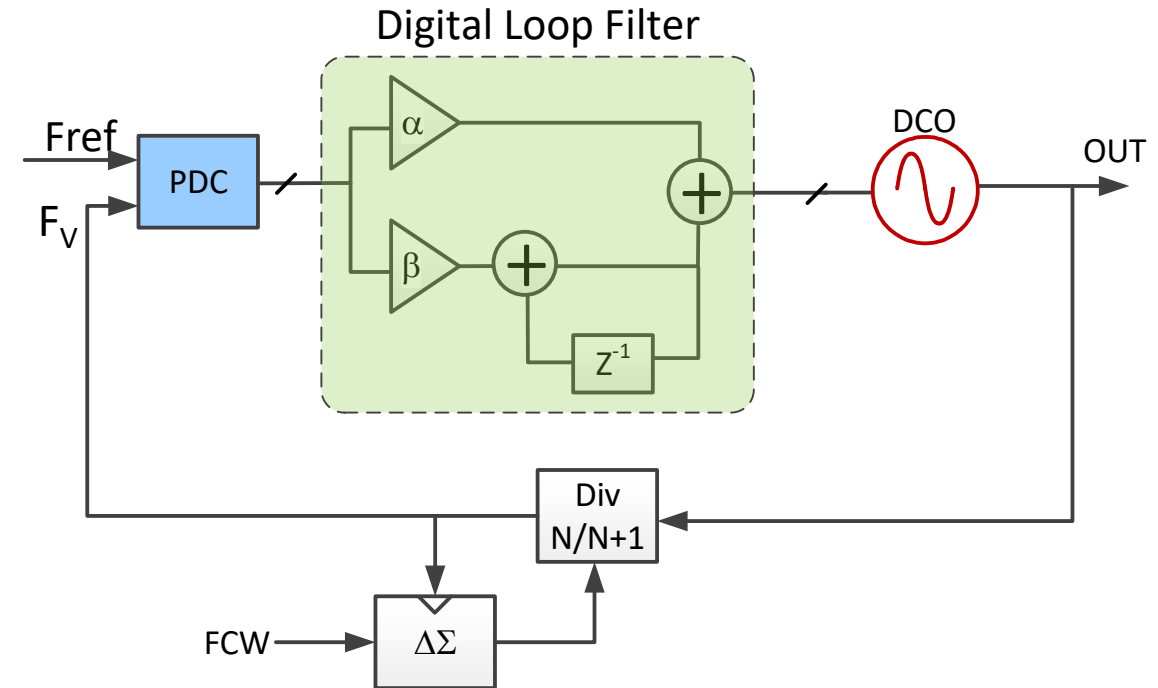
PDC-ADPLL

- Feedforward path
 - PDC replaces PFD in analog PLL
 - PDC quantizes the phase difference between F_{ref} and F_V to digital signal
 - PDC and TDC are the same design
 - TDC measures the absolute time and needs to be calibrated to T_{DCO}
 - PDC measures relative phase and does not need calibration. PDC gain variation only impacts loop gain like charge pump current variation.
- Digital loop filter removes high frequency noise and generates control code for DCO



PDC-ADPLL

- Feedback path
 - N-divider divides DCO output down to F_v , the same frequency as F_{ref} .
 - Delta-Signal modulator controls the instantaneous divide ratio to obtain an average fractional divide ratio.
 - Similar to analog PLL, DSM quantization noise shaping.



Time to Digital Converter (TDC) Overview

- TDC performance specifications
- TDC topologies
 - Delay line based TDC
 - Vernier TDC
 - Ring delay line TDC
 - Coarse-fine TDC with time amplifier
 - Noise shaping TDC
 - ADC based TDC
- TDC quantization noise



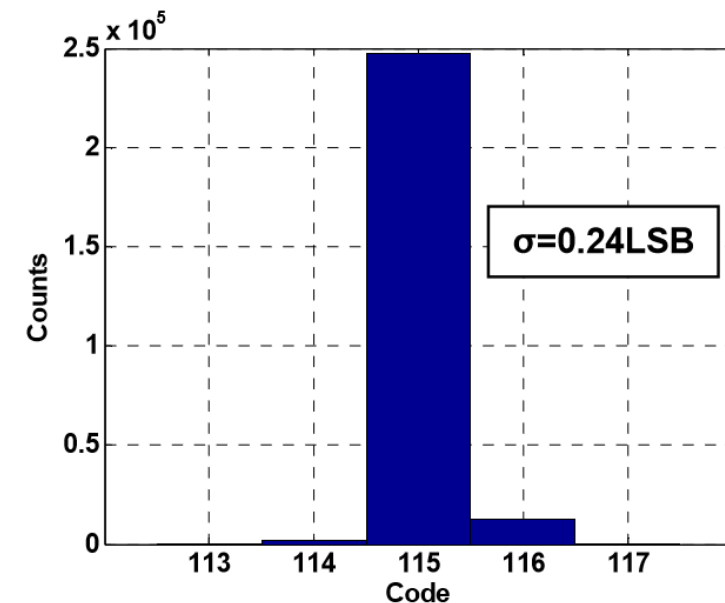
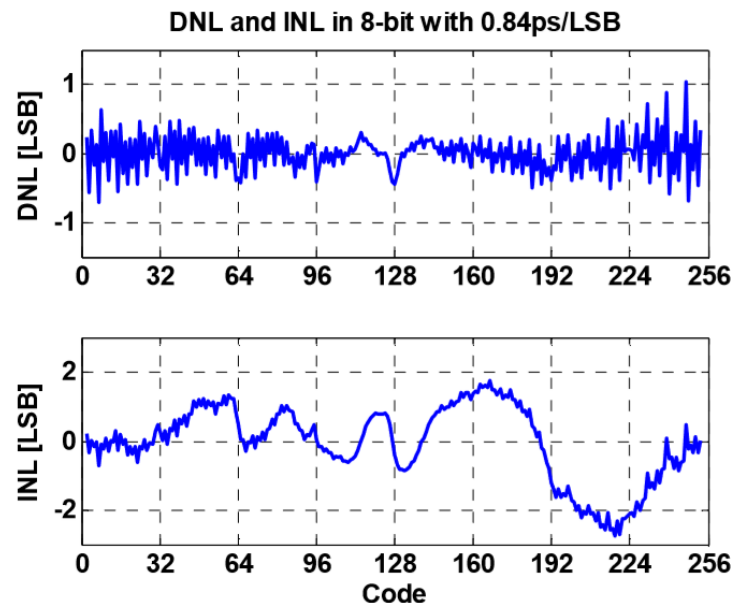
TDC Performance Specifications

- Sampling rate F_s : In ADPLL, the TDC sampling rate equals to the reference frequency F_{ref} .
- Bandwidth: the range of input signal frequencies it can measure. For flash TDCs, it is half of their sampling frequency.
- Raw time resolution T_{res} : LSB quantization resolution. In delay line based TDC, it equals to unit buffer cell propagation delay.
- Effective resolution: actual resolution when circuit noise and mismatch effects are considered
- Full scale range T_{fs} : the maximum time interval that can be measured without saturation or overflow.
- Dynamic range DR: $DR = 10\log_{10} \left(\frac{T_{\text{fs}}}{T_{\text{res}}} \right)$



TDC Performance Specifications

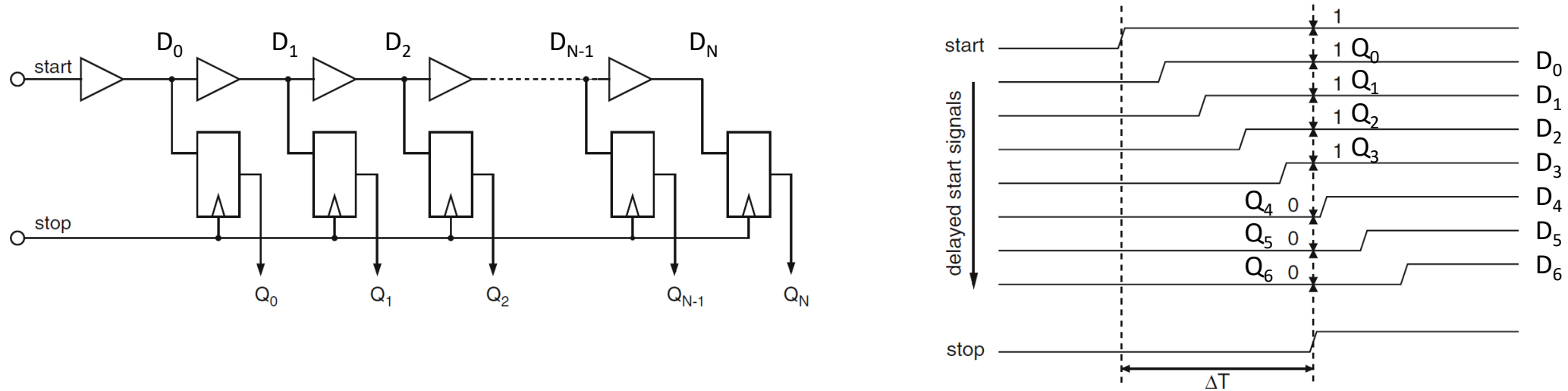
- Linearity: Like analog-to-digital converters, TDC linearity can be characterized by INL and DNL. If there is DSM in the PLL for noise shaping, TDC nonlinearity will cause noise folding to in-band similar to charge pump nonlinearity effect.
- Single-shot precision: measures TDC repeatability. If a constant time interval is measured repeatedly, the digital output values vary with a standard deviation that is called single-shot precision.



Ref [7]



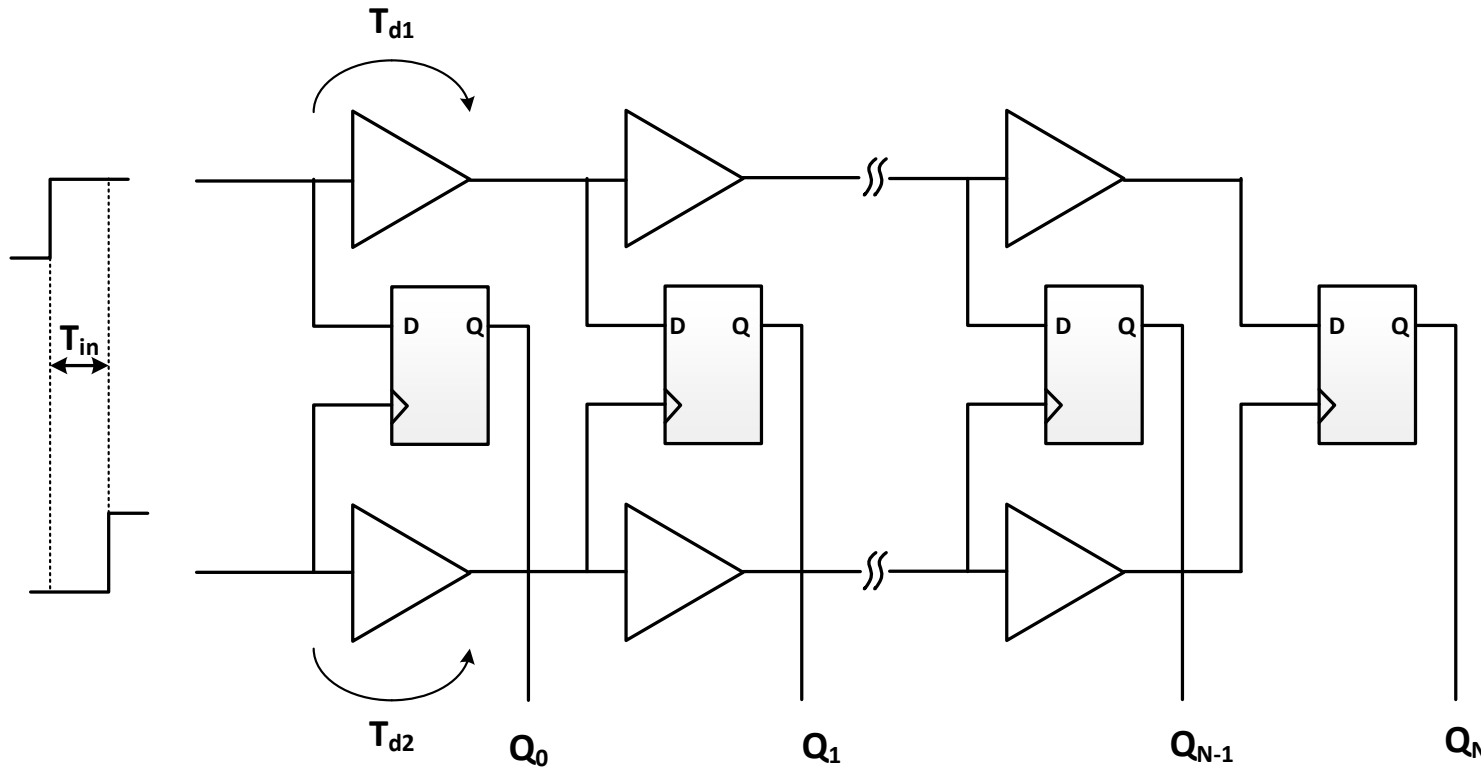
TDC Implementations – Flash TDC



- “Start” signal passes through a delay line that generates sequentially delayed versions, $D_0 \sim D_N$.
- The delayed signals are sampled on the rising edge of the “stop” signal and generate a digital thermometer code latched by the DFFs that represents the time difference between “start” and “stop” signals
- The TDC LSB resolution is one unit delay time τ_d and total range is $N\tau_d$.
- TDC dynamic range is limited by the number of stages.
- LSB is process dependent, 65nm 10~20ps, 28nm 5~10ps, 14nm <5ps



TDC Implementations – Vernier Delay Line TDC

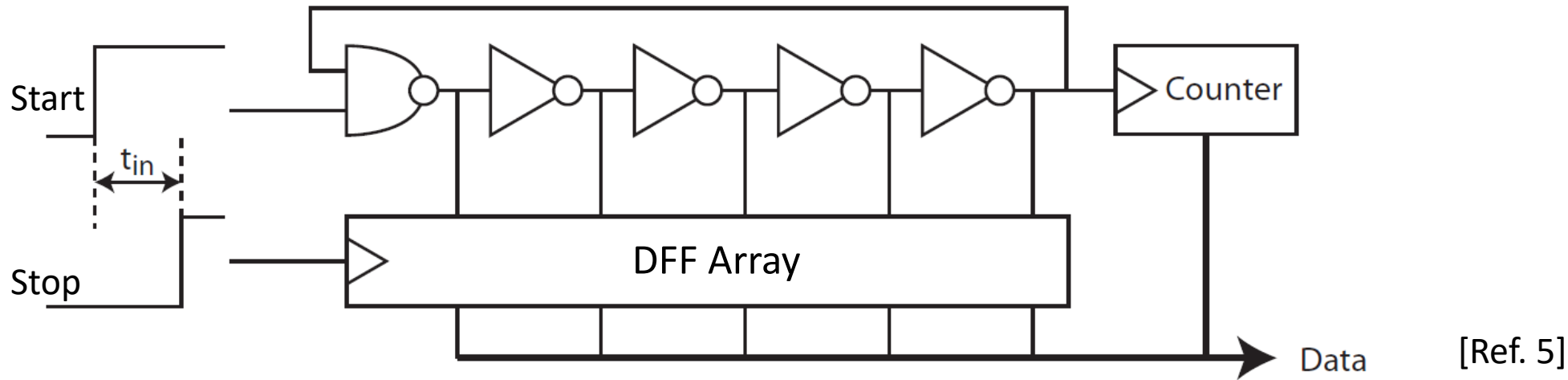


[Ref. 3]

- TDC LSB resolution is determined by one unit buffer propagation delay and limited by the process node.
- Vernier delay line based TDC **improves LSB resolution** by using two delay lines with a unit delay of T_{d1} and T_{d2} ($T_{d1} > T_{d2}$).
- The leading signal is sent to the slower delay line (T_{d1}) and the lagging signal to the faster one (T_{d2}).
- The output is a digital thermometer code whose 0→1 transition happens at the tap where the lagging signal overtakes the leading signal.
- TDC LSB resolution is $T_{d1} - T_{d2}$.
- Flash TDC is a special case of Vernier TDC with $T_{d2} = 0$.



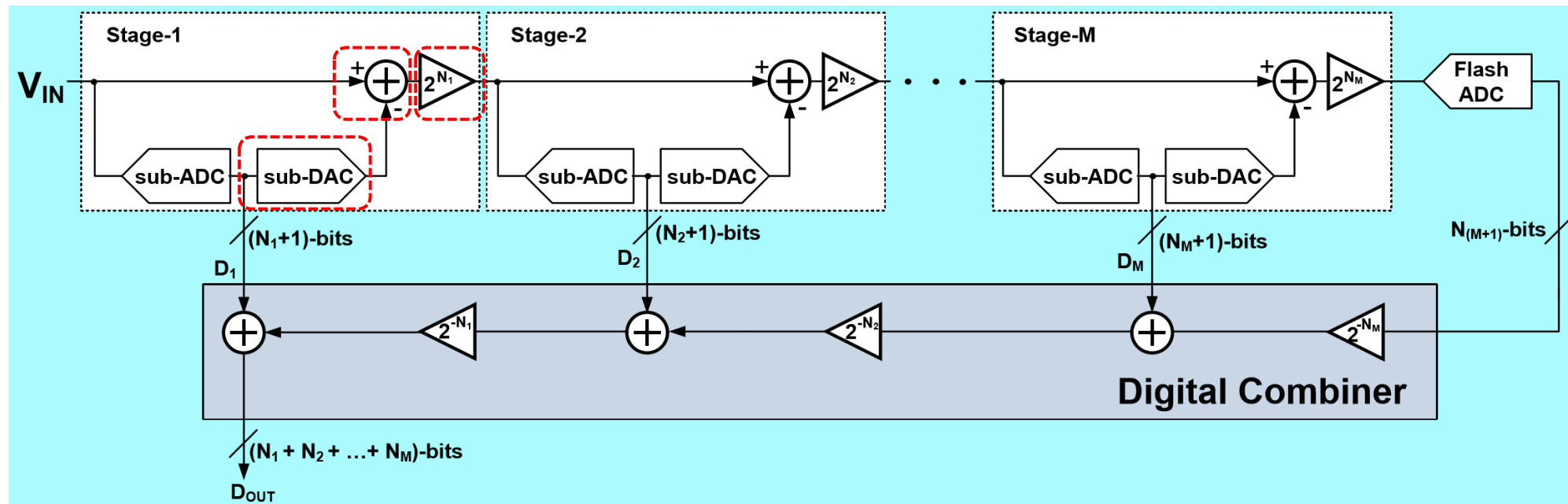
TDC Implementations – Ring Delay Line TDC



- In order to **increase TDC full range** without excessively long delay line, a ring is formed from the delay line to recycle delay cells and a counter is included to count the total cycles the leading edge propagates through the ring.
- The total time $t_{in} = N \cdot \left(\sum_{i=1}^M \tau_i \right) + \sum_{i=1}^k \tau_i$, where M is the total delay cells in the ring, N is the number of cycles "start" signal edge propagates through the ring, and k is the residual number of cells "start" signal edge passes.
- Meta-stability exists when "start" signal borderline completes a cycle and the counter may miscount by 1.
- Combining Vernier delay line with counter ring delay structure for both finer resolution and wide range $\Rightarrow$ Vernier ring TDC [Ref. 6]



Coarse-Fine TDC with Time Amplifier



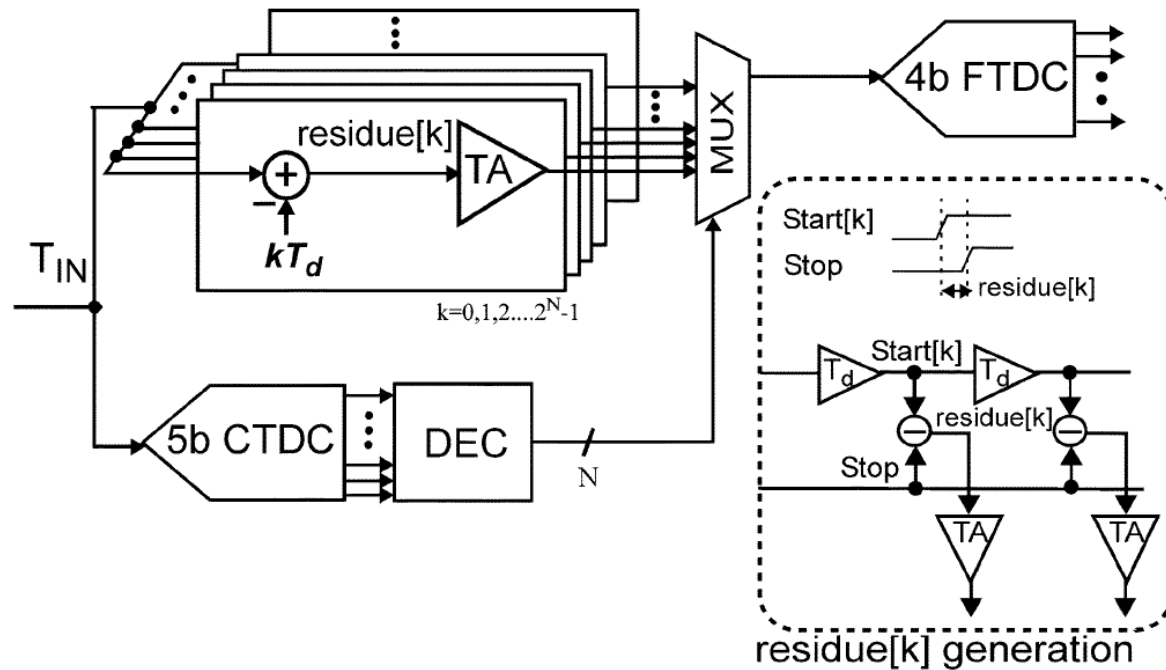
- Pipeline ADC

Ref: B. Sahoo, UIUC

- Each stage resolves a smaller number of bits -> low resolution ADC
- The residual quantization error is amplified as the input to the next stage ADC
- The final output is a combination of the output of each stage with weight factor
- Amplifier gain needs to be calibrated



Coarse-Fine TDC with Time Amplifier

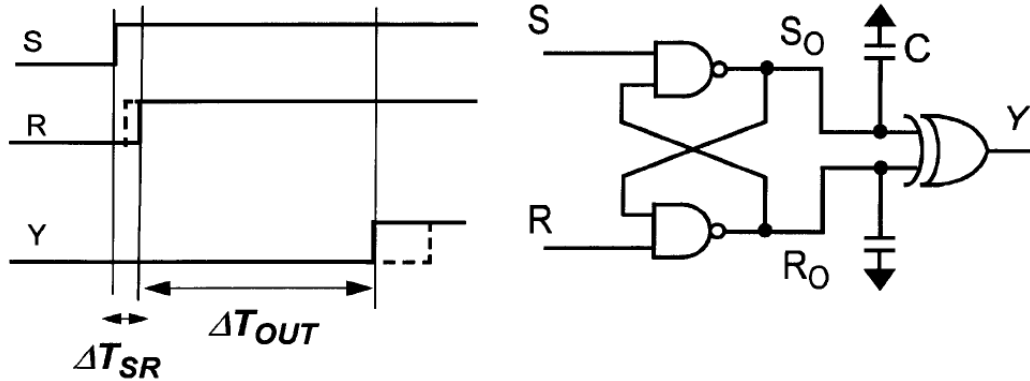


Time residue extraction:
Time DAC
Time subtraction (differential TA)

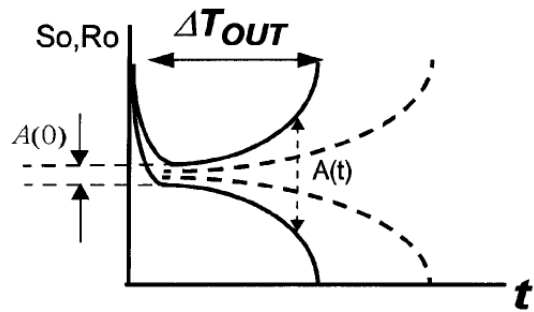
- A coarse-fine TDC improves resolution by amplifying the residue between the input and the closest coarse level and quantizing the amplified residue again with the same coarse resolution. It improves the resolution by the amplification gain.
- Challenges
 - Time residue extraction
 - Time amplification
 - Amplification gain calibration



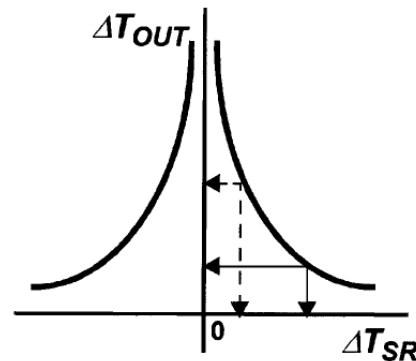
Coarse-Fine TDC with Time Amplifier



(a)



(b)

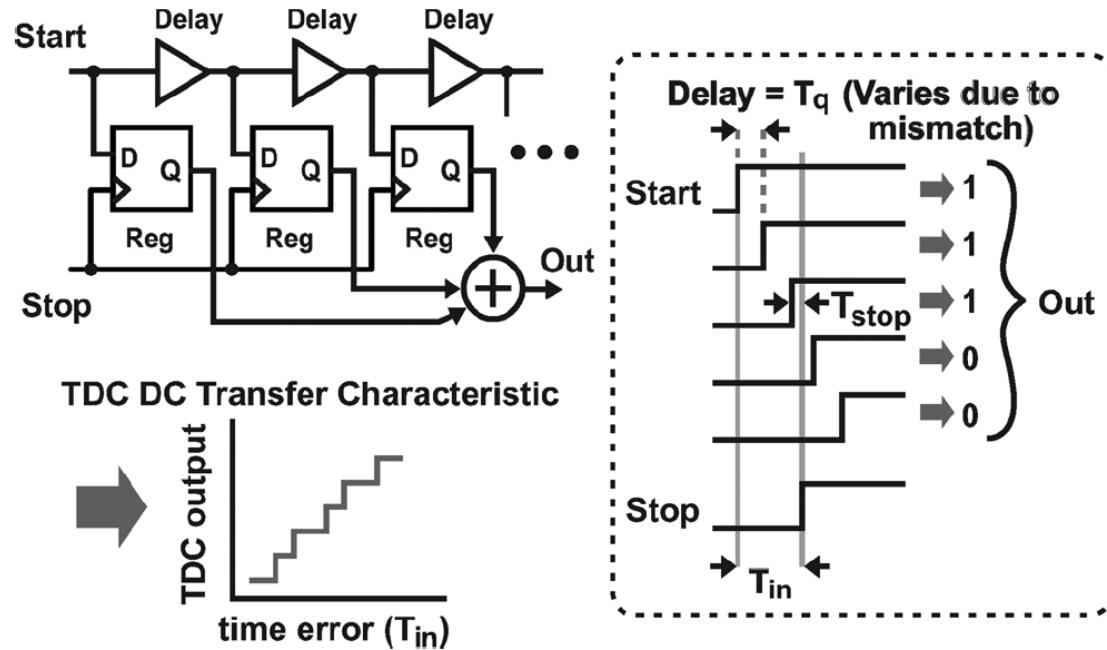


(c)

- Time amplifier is based on SR latch metastability when its input edges are very close.
- ΔT_{SR} is the initial time difference between S and R.
- $A(0) = \alpha \cdot \Delta T_{SR}$, $A(t) = A(0) \cdot e^{t/\tau}$
- Output goes from 0 to 1 when $S_0 - R_0$ reaches XOR threshold voltage V_{TH}
- $\Delta T_{OUT} = \tau(\log V_{TH} - \log |\alpha \cdot \Delta T_{SR}|)$
- τ is the regeneration time constant, an indicator of how fast $S_0 - R_0$ voltage amplitude $A(t)$ builds up for given initial voltage difference
- α is the proportional factor determining the initial voltage difference $A(0)$

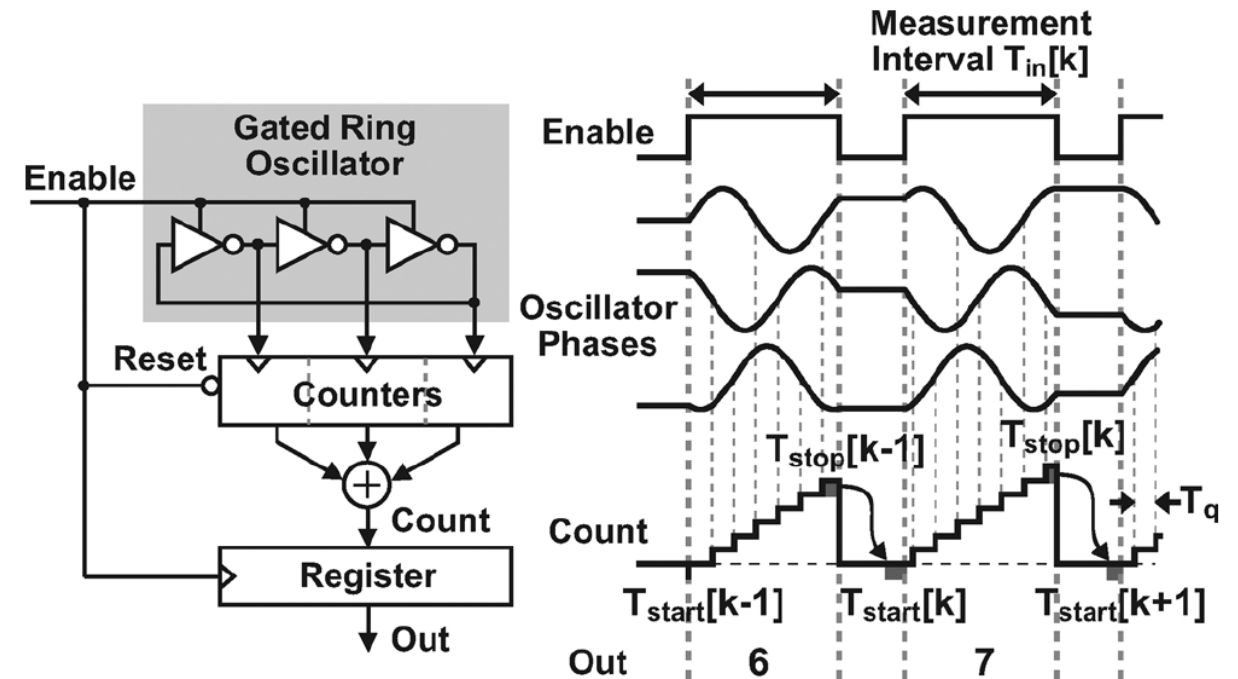


Noise Shaping TDC



Some definitions:

- Input time difference to measure/convert: T_{in}
- TDC resolution: T_q
- Residual time error after $N \cdot T_q$ subtracted: T_{stop}

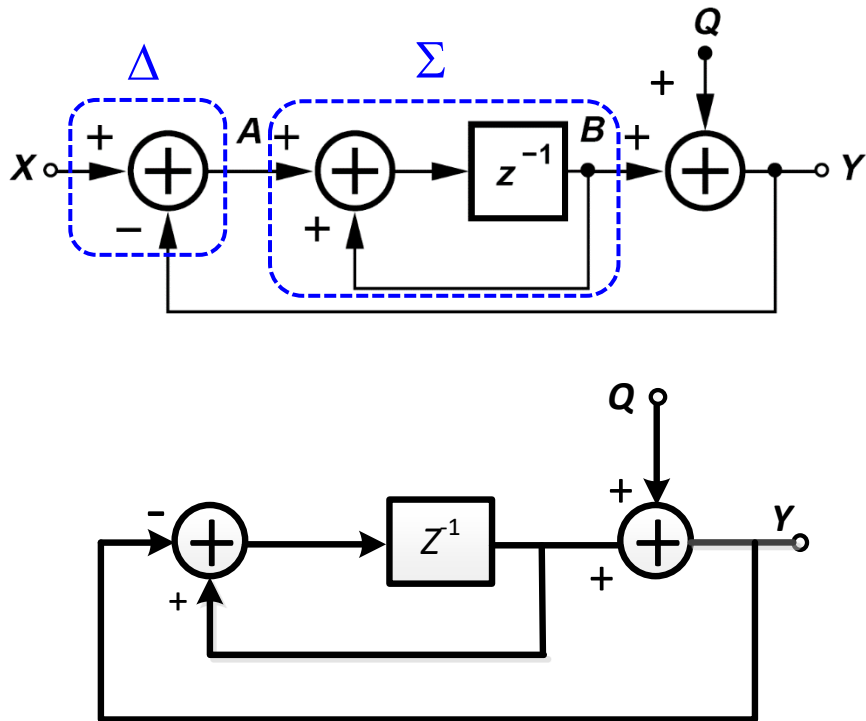


- Gated ring oscillator (GRO) runs during measurement and freezes between measurements.
- Residual quantization time error at the end of (k-1)th measurement interval, $T_{stop}[k-1]$ is transferred to the next measurement interval, $T_{start}[k] = T_{stop}[k-1]$.
- Overall quantization error for each measurement:

$$T_{error}[k] = T_{stop}[k] - T_{start}[k] = T_{stop}[k] - T_{stop}[k-1]$$



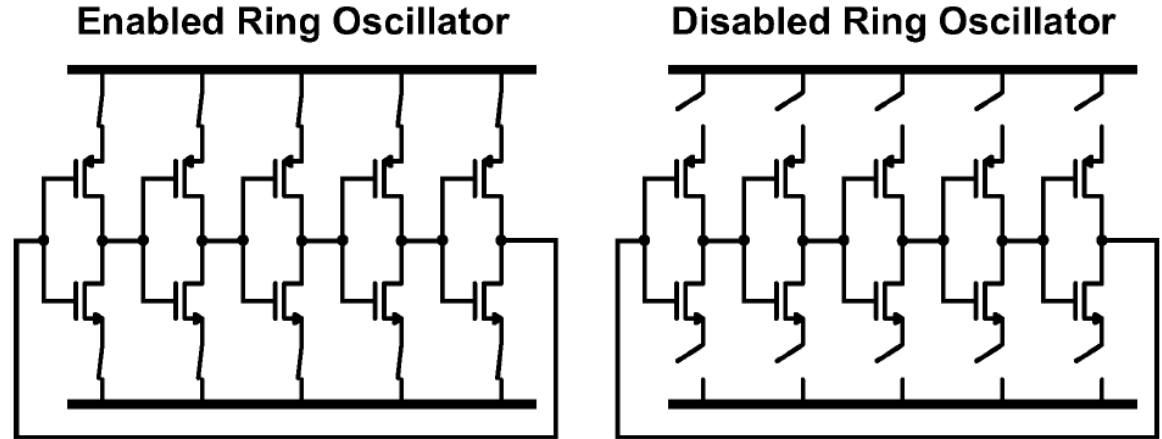
Noise Shaping TDC



Quantization noise TF (high pass): $\frac{Y(Z)}{Q(Z)} = 1 - z^{-1}$

In time domain: $Y[k] = Q[k] - Q[k-1]$

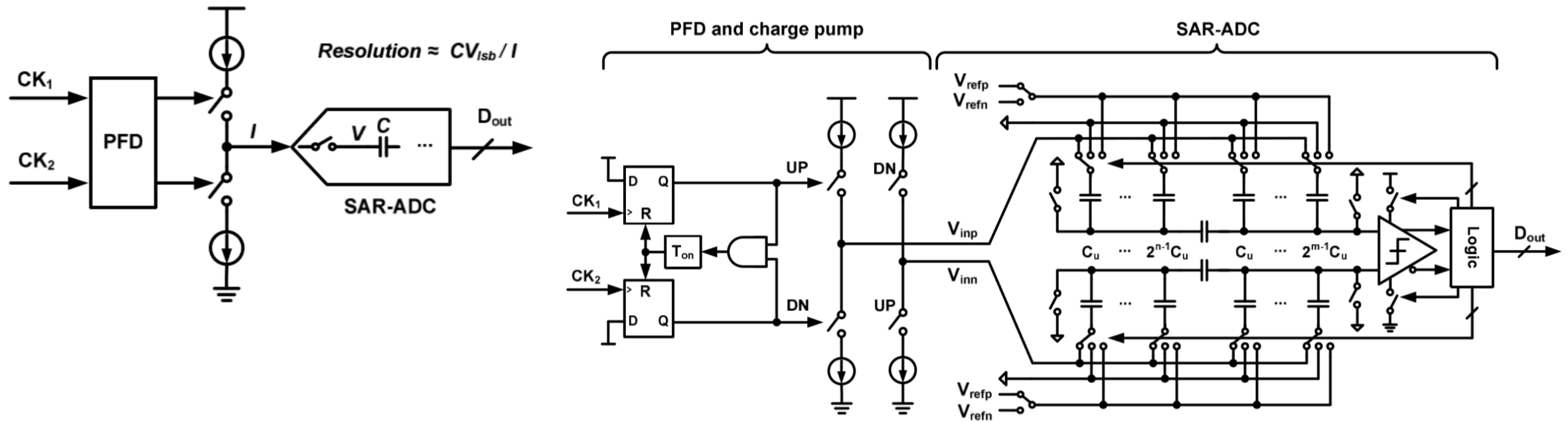
$$T_{\text{error}}[k] = T_{\text{stop}}[k] - T_{\text{stop}}[k-1]$$



- When switches on, RO running
- When switches off, RO freezes
- In freeze state, RO phase information is stored
- Integration and subtraction in time domain
- Other time domain signal processing example with RO:
 - Dual mode ring oscillator (DMRO) frequency to digital converter (Ref [8])



ADC Based TDC



- PFD and charge pump to convert time difference to differential voltage
- SAR ADC to convert the differential voltage to digital output



TDC and ADC Analogy

TDC	ADC
Delay line TDC Vernier delay line TDC Ring delay line TDC	Flash ADC
Coarse-fine TDC with time amplifier [Ref 4]	Pipeline ADC
Gated ring oscillator TDC [Ref 7]	Delta-sigma ADC
SAR TDC	SAR ADC
ADC based TDC	Time based ADC VCO based ADC



TDC Quantization Noise

- Assume TDC resolution is T_{res} , the variance of quantization error is

$$\sigma_t^2 = \frac{T_{res}^2}{12}$$

- The quantization error in phase (*rad*) is the standard deviation of quantization error normalized to the unit interval (DCO period) and multiplied by 2π .

$$\sigma_\phi = 2\pi \frac{\sigma_t}{T_V}$$

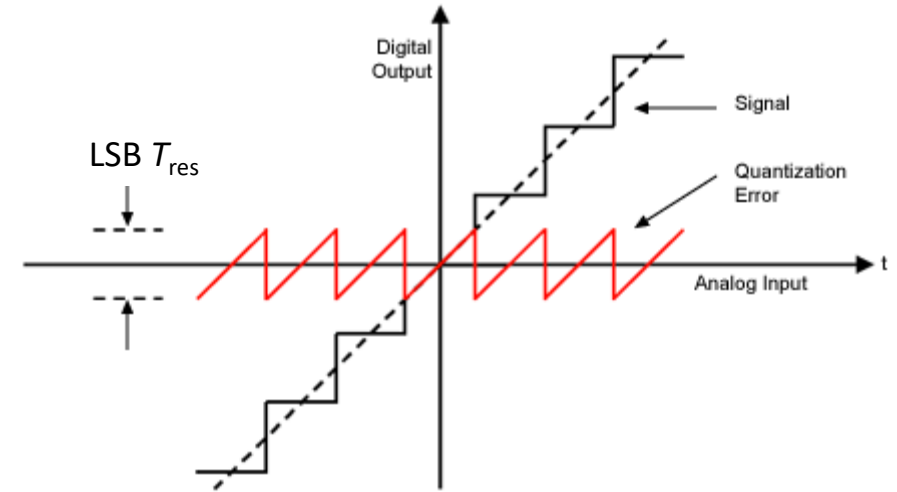
- The total phase noise power is uniformly distributed from dc to the Nyquist frequency f_R , the single-sided spectral density is

$$\mathcal{L} = \frac{\sigma_\phi^2}{f_R} = \frac{(2\pi)^2}{12} \left(\frac{T_{res}}{T_V} \right)^2 \frac{1}{f_R}$$

- Example: $T_{res}=10\text{ps}$, $f_v=5\text{GHz}$, $T_v=200\text{ps}$, $f_R=40\text{MHz} \Rightarrow$

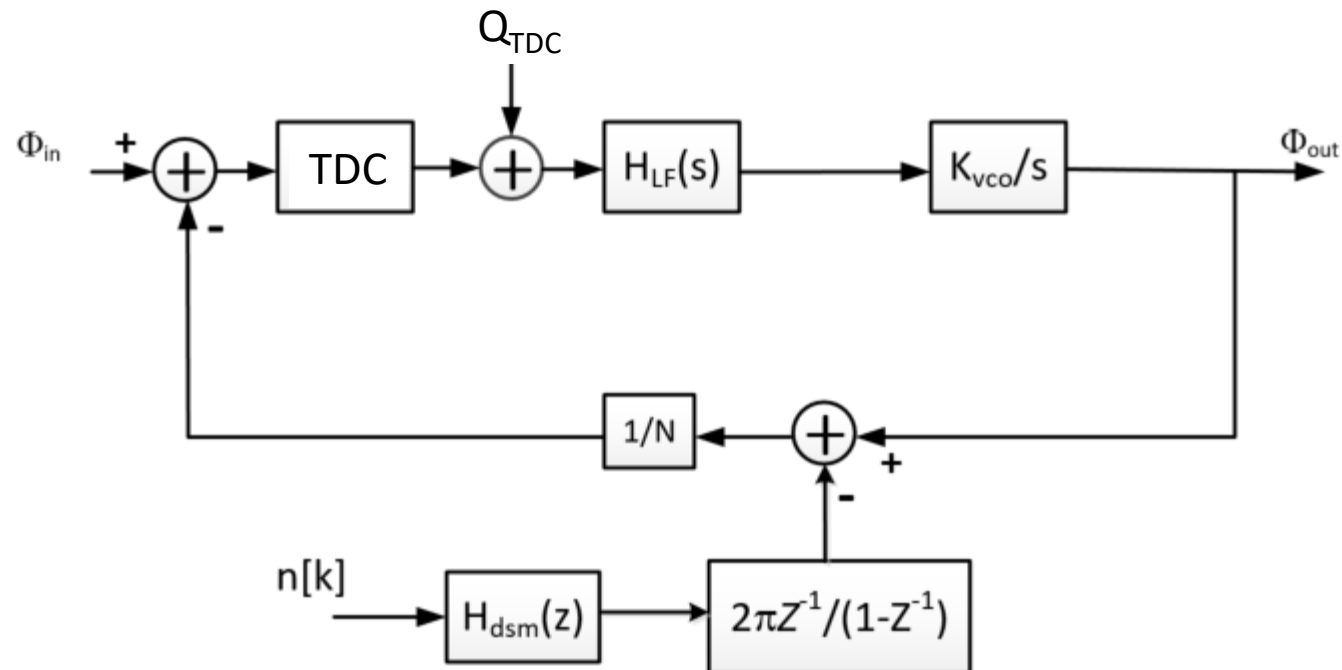
$$\mathcal{L}_{dB} = 10 \log \left(\frac{(2\pi)^2}{12} \left(\frac{10^{-11}}{2 \cdot 10^{-10}} \right)^2 \frac{1}{4 \cdot 10^7} \right) = -96.9 \text{ dBc/Hz}$$

This performance is not sufficient for high QAM modulation, such as 256QAM or 1024QAM.



TDC Quantization Noise

- As one of the noise sources in the feedforward path, similar to charge pump noise in analog PLL, TDC quantization noise is low pass filtered by the loop filter.



ADPLL References

- 1) Robert B. Staszewski and Poras T. Balsara, All-Digital Frequency Synthesizer in Deep-Submicron CMOS, Wiley, 2006.
- 2) Stephan Henzler, Time-to-Digital Converters, Springer, 2010.
- 3) P. Dudek, S. Szczepanski and J. V. Hatfield, “A high-resolution CMOS time-to-digital converter utilizing a Vernier delay line,” pp. 240-247, JSSC 2000.
- 4) Minjae Lee and Asad A. Abidi, “A 9b, 1.25ps resolution coarse-fine time-to-digital converter in 90nm CMOS that amplifies a time residue,” pp. 769-777, JSSC 2008.
- 5) <https://essay.utwente.nl/69501/1/MSc.%20report%20Broek%20van%20den,%20J.D.A.pdf>



TDC References

- 1) Robert B. Staszewski and Poras T. Balsara, All-Digital Frequency Synthesizer in Deep-Submicron CMOS, Wiley, 2006.
- 2) Stephan Henzler, Time-to-Digital Converters, Springer, 2010.
- 3) P. Dudek, S. Szczepanski and J. V. Hatfield, "A high-resolution CMOS time-to-digital converter utilizing a Vernier delay line," pp. 240-247, JSSC 2000.
- 4) Minjae Lee and Asad A. Abidi, "A 9b, 1.25ps resolution coarse-fine time-to-digital converter in 90nm CMOS that amplifies a time residue," pp. 769-777, JSSC 2008.
- 5) <https://essay.utwente.nl/69501/1/MSc.%20report%20Broek%20van%20den,%20J.D.A.pdf>
- 6) Z. Xu, et. al., A 0.84ps-LSB 2.47mW time-to-digital converter using charge pump and SAR-ADC, CICC 2013
- 7) M. Straayer and M. Perrott, A multi-path gated ring oscillator TDC with first-order noise shaping, JSSC, 2009
- 8) C. Wu, G. Zhao and I. Galton, A 3.5GHz digital fractional-N PLL frequency synthesizer based on ring oscillator frequency-to-digital conversion, JSSC, 2015
- 9) V. Kratyuk, et. al., A digital PLL with a stochastic time-to-digital converter, TCAS-1, 2009
- 10) M. Lee and A. A. Abidi, A 9 b, 1.25 ps Resolution Coarse–Fine Time-to-Digital Converter in 90nm CMOS that Amplifies a Time Residue, JSSC 2008

